Eigenvalues and Eigenvectors, $\vec{\S}$ 5.4

Recall from last class:

- **Theorem:** Let $\mathcal{B} = \{b_1, \ldots, b_n\}$ and $\mathcal{C} = \{c_1, \ldots, c_n\}$ be bases of a vector space V. Then there exists a unique $n \times n$ matrix P such that $[x]_{\mathcal{C}} = P[x]_{\mathcal{B}}$.
- We call P the **change of coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$**. In general

  $$P = [[b_1]_{\mathcal{C}} \ \ldots \ [b_n]_{\mathcal{C}}].$$

  Also P^{-1} is the change of coordinates matrix from $\mathcal{C}$ to $\mathcal{B}$.

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1. Let V be a vector space with basis $\mathcal{B} = \{b_1, \ldots, b_n\}$ and W be a vector space with basis $\mathcal{C} = \{c_1, \ldots, c_m\}$. Given a linear transformation $T : V \to W$, we naturally seek an associated matrix to represent T using the bases; namely we want an $m \times n$ matrix A such that $[T(x)]_{\mathcal{C}} = A[x]_{\mathcal{B}}$ for all $x \in V$. How can we find the matrix A?

We call A the **matrix of T relative to bases $\mathcal{B}$ and $\mathcal{C}$**. In general $A = [[T(b_1)]_{\mathcal{C}} \ \ldots \ [T(b_n)]_{\mathcal{C}}]$.

2. Let $\mathcal{B} = \{b_1, b_2, b_3\}$ and $\mathcal{D} = \{d_1, d_2\}$ be bases for V and W respectively. Let $T : V \to W$ be the linear transformation such that $T(b_1) = 3d_1 - 5d_2$, $T(b_2) = -d_1 + 6d_2$, and $T(b_3) = 4d_2$. Find the matrix of T relative to $\mathcal{B}$ and $\mathcal{D}$.

3. Let $T : \mathbb{P}_2 \to \mathbb{P}_2$ by $T(p) = \frac{dp}{dt}$. Find the matrix of T relative to the standard basis.

4. Let $A = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix}$. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ given by $T(x) = Ax$ and $\mathcal{B} = \left\{ \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right\}$.

(a) Find the matrix of T relative to $\mathcal{B}$.

(b) Find A^{2019}.

Diagonal Representation Theorem: Suppose $A = PDP^{-1}$ where D is a diagonal matrix. If $\mathcal{B}$ is the basis for $\mathbb{R}^n$ formed by the columns of P, then D is the matrix of T relative to $\mathcal{B}$.

5. Prove the theorem.

Definition 1. *If A and B are $n \times n$ matrices, then A is* **similar** *to B if there exists an invertible matrix P such that $A = PBP^{-1}$.*

Definition 2. *If A is an $n \times n$ matrix, then $\lambda \in \mathbb{R}$ is an* **eigenvalue** *of A if there exists a non-zero vector v, called an* **eigenvector** *corresponding to λ, such that*

$$Av = \lambda v.$$